

MEMBUAT HAK AKSES

Tahap 1: Membuat Middleware untuk Hak Akses (Role Middleware)

Di Laravel, middleware bertindak sebagai penjaga gerbang (gatekeeper) sebelum user bisa mengakses sebuah Route atau Controller tertentu.

1. Membuat File Middleware:

Jalankan perintah berikut via terminal Docker untuk membuat middleware kustom:

```
fajaroktaviannugraha@MacBook-Air-Fajar project-peminjaman-alat % docker exec -it alat_app php artisan make:middleware CheckRole
INFO: Middleware [app/Http/Middleware/CheckRole.php] created successfully.
```

2. Menulis Logika Middleware (**app/Http/Middleware/CheckRole.php**):

```
public function handle(Request $request, Closure $next, ...$roles): Response
{
    // Cek apakah user sudah login dan apakah rolenya ada di dalam parameter yang diizinkan
    if (!auth()->check() || !in_array(auth()->user()->role, $roles)) {
        abort(403, 'Unauthorized action.');
```

3. Mendaftarkan Middleware di bootstrap/app.php:

Pada Laravel versi terbaru, daftarkan alias middleware di file bootstrap/app.php:

```
src > bootstrap > app.php
1  <?php
2
3  use Illuminate\Foundation\Application;
4  use Illuminate\Foundation\Configuration\Exceptions;
5  use Illuminate\Foundation\Configuration\Middleware;
6
7  return Application::configure(basePath: dirname(__DIR__))
8      ->withRouting(
9          web: __DIR__.'/../routes/web.php',
10         commands: __DIR__.'/../routes/console.php',
11         health: '/up',
12     )
13     ->withMiddleware(function (Middleware $middleware): void {
14         $middleware->alias([
15             'role' => \App\Http\Middleware\CheckRole::class,
16         ]);
17     })
18     ->withExceptions(function (Exceptions $exceptions): void {
19         //
20     })->create();
```

Tahap 2: Membuat Controller Fungsional

Berikut adalah rancangan Controller dasar untuk masing-masing role (Admin, Petugas, dan Peminjam) beserta contoh logika fungsional untuk menangani fitur utama seperti CRUD alat, persetujuan pinjaman, dan pengajuan pinjam.

1. Membuat File Controller via Terminal Docker

Jalankan perintah berikut untuk membuat ketiga controller tersebut:

```
fajaroktaviannugraha@MacBook-Air-Fajar project-peminjaman-alat % docker exec -it alat_app php artisan make:controller AdminController
INFO Controller [app/Http/Controllers/AdminController.php] created successfully.
fajaroktaviannugraha@MacBook-Air-Fajar project-peminjaman-alat % docker exec -it alat_app php artisan make:controller PetugasController
INFO Controller [app/Http/Controllers/PetugasController.php] created successfully.
fajaroktaviannugraha@MacBook-Air-Fajar project-peminjaman-alat % docker exec -it alat_app php artisan make:controller PeminjamController
INFO Controller [app/Http/Controllers/PeminjamController.php] created successfully.
```

File ada di app/Http/Controllers

```

└─ app
  └─ Http
    └─ Controllers
      ├── AdminController.php
      ├── AuthController.php
      ├── Controller.php
      ├── PeminjamController.php
      └── PetugasController.php
```

2. AdminController.php

Admin memegang kendali penuh (CRUD User, CRUD Alat, CRUD Kategori, dan melihat Log Aktivitas). Berikut contoh logika untuk pengelolaan Alat dan Kategori:

```

src > app > Http > Controllers > AdminController.php > {} App\Http\Controllers > AdminController
1  <?php
2
3  namespace App\Http\Controllers;
4
5  use App\Models\Alat;
6  use App\Models\Kategori;
7  use App\Models\User;
8  use App\Models\LogAktivitas;
9  use Illuminate\Http\Request;
10
11  5 references | 0 implementations
12  class AdminController extends Controller
13  {
14      // Menampilkan Dashboard Admin & Log Aktivitas
15      1 reference | 0 overrides
16      public function index()
17      {
18          $logs = LogAktivitas::with('user')->latest()->take(10)->get();
19          return view('admin.dashboard', compact('logs'));
20      }
21
22      // CRUD Alat: Menampilkan daftar alat
23      1 reference | 0 overrides
24      public function indexAlat()
25      {
26          $alats = Alat::with('kategori')->get();
27          return view('admin.alat.index', compact('alats'));
28      }
29

```

```

27      // Menyimpan Alat Baru
28      1 reference | 0 overrides
29      public function storeAlat(Request $request)
30      {
31          $request->validate([
32              'kategori_id' => 'required',
33              'nama_alat' => 'required|string|max:255',
34              'stok' => 'required|integer',
35              'status_kondisi' => 'required|string',
36          ]);
37
38          Alat::create($request->all());
39
40          // Catat Log Aktivitas
41          LogAktivitas::create([
42              'user_id' => auth()->id(),
43              'aktivitas' => 'Menambahkan alat baru: ' . $request->nama_alat
44          ]);
45
46          return redirect()->back()->with('success', 'Alat berhasil ditambahkan.');
```

```

47
48      // CRUD User (Manajemen User Admin, Petugas, Peminjam)
49      1 reference | 0 overrides
50      public function indexUser()
51      {
52          $users = User::all();
53          return view('admin.user.index', compact('users'));
54      }
55

```

3. PetugasController.php

Petugas bertugas memverifikasi peminjaman, memantau pengembalian alat, serta mencatat denda/kondisi saat alat kembali.

```

src > app > Http > Controllers > PetugasController.php > {} App\Http\Controllers > PetugasController > indexPeminjaman()
1  <?php
2
3  namespace App\Http\Controllers;
4
5  use App\Models\Peminjaman;
6  use App\Models\Pengembalian;
7  use App\Models\Alat;
8  use Illuminate\Http\Request;
9  use Illuminate\Support\Facades\DB;
10
11  4 references | 0 implementations
12  class PetugasController extends Controller
13  {
14      // Menampilkan daftar pengajuan peminjaman dari siswa/peminjam
15      1 reference | 0 overrides
16      public function indexPeminjaman()
17      {
18          $peminjamans = Peminjaman::with(['user', 'detailPinjams.alat'])->latest()->get();
19          return view('petugas.peminjaman.index', compact('peminjamans'));
20      }

```

```

21  public function setujuPeminjaman($id)
22  {
23      DB::beginTransaction();
24      try {
25          $peminjaman = Peminjaman::with('detailPinjams')->findOrFail($id);
26          $peminjaman->update(['status' => 'dipinjam']);
27
28          // Kurangi stok alat secara otomatis
29          foreach ($peminjaman->detailPinjams as $detail) {
30              $alat = Alat::findOrFail($detail->alat_id);
31              $alat->stok -= $detail->jumlah;
32              $alat->save();
33          }
34
35          DB::commit();
36          return redirect()->back()->with('success', 'Peminjaman disetujui dan stok alat dikurangi.');
```

```

37      } catch (\Exception $e) {
38          DB::rollback();
39          return redirect()->back()->with('error', 'Terjadi kesalahan: ' . $e->getMessage());
40      }
41  }
42

```

```

public function prosesPengembalian(Request $request, $peminjamanId)
{
    $request->validate([
        'kondisi_kembali' => 'required|string',
        'denda' => 'nullable|integer',
    ]);

    DB::beginTransaction();
    try {
        $peminjaman = Peminjaman::with('detailPinjams')->findOrFail($peminjamanId);

        // Simpan data pengembalian
        Pengembalian::create([
            'peminjaman_id' => $peminjaman->id,
            'tgl_kembali' => now(),
            'kondisi_kembali' => $request->kondisi_kembali,
            'denda' => $request->denda ?? 0,
            'petugas_id' => auth()->id(),
        ]);

        // Update status peminjaman jadi selesai
        $peminjaman->update(['status' => 'selesai']);

        // Kembalikan stok alat ke inventaris
        foreach ($peminjaman->detailPinjams as $detail) {
            $alat = Alat::findOrFail($detail->alat_id);
            $alat->stok += $detail->jumlah;
            $alat->save();
        }

        DB::commit();
        return redirect()->back()->with('success', 'Pengembalian berhasil dicatat dan stok dipulihkan.');
```

```

    } catch (\Exception $e) {
        DB::rollback();
        return redirect()->back()->with('error', 'Terjadi kesalahan: ' . $e->getMessage());
    }
}

```

4. PeminjamController.php

Peminjam (Siswa/Guru) dapat melihat katalog alat, mengajukan peminjaman, dan melihat riwayat pinjaman mereka sendiri.

```

src > app > Http > Controllers > PeminjamController.php > {} App\Http\Controllers > PeminjamController > ajukanPeminjaman()
1  <?php
2
3  namespace App\Http\Controllers;
4
5  use App\Models\Alat;
6  use App\Models\Peminjaman;
7  use App\Models\DetailPinjam;
8  use Illuminate\Http\Request;
9  use Illuminate\Support\Facades\DB;
10
11  4 references | 0 implementations
12  class PeminjamController extends Controller
13  {
14      // Melihat daftar/katalog alat yang tersedia
15      1 reference | 0 overrides
16      public function katalogAlat()
17      {
18          $alats = Alat::with('kategori')->where('stok', '>', 0)->get();
19          return view('peminjam.katalog', compact('alats'));
20      }
21  }

```

```

21 public function ajukanPeminjaman(Request $request)
22 {
23     $request->validate([
24         'tgl_kembali_plan' => 'required|date|after:today',
25         'alat_id' => 'required|array',
26         'jumlah' => 'required|array',
27     ]);
28
29     DB::beginTransaction();
30     try {
31         // Buat header peminjaman
32         $peminjaman = Peminjaman::create([
33             'user_id' => auth()->id(),
34             'tgl_pinjam' => now(),
35             'tgl_kembali_plan' => $request->tgl_kembali_plan,
36             'status' => 'diajukan',
37         ]);
38
39         // Masukkan daftar alat yang dipinjam ke detail_pinjam
40         foreach ($request->alat_id as $index => $alatId) {
41             DetailPinjam::create([
42                 'peminjaman_id' => $peminjaman->id,
43                 'alat_id' => $alatId,
44                 'jumlah' => $request->jumlah[$index],
45             ]);
46         }
47
48         DB::commit();
49         return redirect()->route('peminjam.riwayat')->with('success', 'Pengajuan peminjaman berhasil dikirim.');
```

```

56 // Melihat riwayat peminjaman user yang sedang login
57 1 reference | 0 overrides
58 public function riwayatPeminjaman()
59 {
60     $peminjamans = Peminjaman::with('detailPinjams.alat')
61         ->where('user_id', auth()->id())
62         ->latest()
63         ->get();
64
65     return view('peminjam.riwayat', compact('peminjamans'));
66 }
67
```

Tahap 3: Menerapkan Middleware pada Route (**routes/web.php**)

Setelah middleware siap, kita bisa langsung mengamankan jalur akses berdasarkan matriks fitur yang kita buat (Admin, Petugas, Peminjam):

1. Konfigurasi routes/web.php

```
src > routes > web.php
```

```
1  <?php
2
3  use Illuminate\Support\Facades\Route;
4  use App\Http\Controllers\AdminController;
5  use App\Http\Controllers\PetugasController;
6  use App\Http\Controllers\PeminjamController;
7  use App\Http\Controllers\AuthController;
8
9  Route::get('/', function () {
10     return view('welcome');
11 });
12
13 //admin
14 Route::middleware(['auth', 'role:admin'])->prefix('admin')->name('admin.')->group(function () {
15     Route::get('/dashboard', [AdminController::class, 'index'])->name('dashboard');
16
17     // CRUD Alat
18     Route::get('/alat', [AdminController::class, 'indexAlat'])->name('alat.index');
19     Route::post('/alat', [AdminController::class, 'storeAlat'])->name('alat.store');
20
21     // CRUD User
22     Route::get('/users', [AdminController::class, 'indexUser'])->name('user.index');
23 });
```

```
25 //petugas
26 Route::middleware(['auth', 'role:petugas,admin'])->prefix('petugas')->name('petugas.')->group(function () {
27     // Peminjaman & Persetujuan
28     Route::get('/peminjaman', [PetugasController::class, 'indexPeminjaman'])->name('peminjaman.index');
29     Route::post('/peminjaman/{id}/setujui', [PetugasController::class, 'setujuiPeminjaman'])->name('peminjaman.setujui');
30
31     // Pengembalian & Denda
32     Route::post('/pengembalian/{id}', [PetugasController::class, 'prosesPengembalian'])->name('pengembalian.proses');
33 });
34
35 //peminjam
36 Route::middleware(['auth', 'role:peminjam'])->prefix('peminjam')->name('peminjam.')->group(function () {
37     // Katalog & Pengajuan
38     Route::get('/katalog', [PeminjamController::class, 'katalogAlat'])->name('katalog');
39     Route::post('/peminjaman/ajukan', [PeminjamController::class, 'ajukanPeminjaman'])->name('peminjaman.ajukan');
40     Route::get('/riwayat', [PeminjamController::class, 'riwayatPeminjaman'])->name('riwayat');
41 });
42
43 // Route Tamu (Belum Login)
44 Route::middleware('guest')->group(function () {
45     Route::get('/login', [AuthController::class, 'showLoginForm'])->name('login');
46     Route::post('/login', [AuthController::class, 'login']);
47 });
48
49 // Route Logout (Harus sudah login)
50 Route::post('/logout', [AuthController::class, 'logout'])->name('logout')->middleware('auth');
```

Tahap 4: Membuat AuthController

1. Membuat AuthController

Buat controller khusus untuk menangani proses login dan logout:

```
• fajaroktaviannugraha@MacBook-Air-Fajar project-peminjaman-alat % docker exec -it alat_app php artisan make:controller AuthController
INFO Controller [app/Http/Controllers/AuthController.php] created successfully.
```

Isi file app/Http/Controllers/AuthController.php:

```
src > app > Http > Controllers > AuthController.php > {} App\Http\Controllers > AuthController
1  <?php
2
3  namespace App\Http\Controllers;
4
5  use Illuminate\Http\Request;
6  use Illuminate\Support\Facades\Auth;
7
8  4 references | 0 implementations
9  class AuthController extends Controller
10 {
11     // Menampilkan Form Login
12     1 reference | 0 overrides
13     public function showLoginForm()
14     {
15         return view('auth.login');
16     }
17
18     // Memproses Login
19     1 reference | 0 overrides
20     public function login(Request $request)
21     {
22         $credentials = $request->validate([
23             'email' => ['required', 'email'],
24             'password' => ['required'],
25         ]);
26
27         if (Auth::attempt($credentials)) {
28             $request->session()->regenerate();
29
30             $user = Auth::user();
31
32             // Redirect berdasarkan Role sesuai matriks Anda
33             if ($user->role === 'admin') {
34                 return redirect()->route('admin.dashboard');
35             } elseif ($user->role === 'petugas') {
36                 return redirect()->route('petugas.peminjaman.index');
37             } elseif ($user->role === 'peminjam') {
38                 return redirect()->route('peminjam.katalog');
39             }
40
41             Auth::logout();
42             return redirect()->route('login')->with('error', 'Role tidak dikenali.');
```

2. Membuat Tampilan Form Login (resources/views/auth/login.blade.php)

Buat folder auth di dalam resources/views/, lalu buat file login.blade.php. Kita gunakan HTML biasa yang bersih dan simpel:

```
src > resources > views > auth > login.blade.php
1  <!DOCTYPE html>
2  <html lang="id">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>Login - Sistem Peminjaman Alat</title>
7      <!-- Memuat Tailwind CSS melalui CDN -->
8      <script src="https://cdn.tailwindcss.com"></script>
9  </head>
10 <body class="bg-gray-100 flex items-center justify-center h-screen">
11
12     <!-- Card Container -->
13     <div class="bg-white p-8 rounded-lg shadow-md w-96">
14         <h3 class="text-2xl font-bold text-center text-gray-800 mb-6">Login Sistem</h3>
15
16         <!-- Alert Error Session -->
17         @if(session('error'))
18             <div class="mb-4 p-3 bg-red-100 border border-red-400 text-red-700 rounded text-sm">
19                 {{ session('error') }}
20             </div>
21         @endif
22
23         <!-- Alert Error Validasi -->
24         @if($errors->any())
25             <div class="mb-4 p-3 bg-red-100 border border-red-400 text-red-700 rounded text-sm">
26                 <ul class="list-disc pl-5 mb-0">
27                     @foreach($errors->all() as $error)
28                         <li>{{ $error }}</li>
29                     @endforeach
30                 </ul>
31             </div>
32         @endif
33
34         <form action="{{ route('login') }}" method="POST">
35             @csrf
36
37             <!-- Input Email -->
38             <div class="mb-4">
39                 <label class="block text-gray-700 text-sm font-semibold mb-2">Email</label>
40                 <input type="email" name="email" value="{{ old('email') }}" required
41                     class="w-full px-3 py-2 border border-gray-300 rounded-lg focus:outline-none focus:ring-2 focus:ring-blue-500">
42             </div>
43
44             <!-- Input Password -->
45             <div class="mb-6">
46                 <label class="block text-gray-700 text-sm font-semibold mb-2">Password</label>
47                 <input type="password" name="password" required
48                     class="w-full px-3 py-2 border border-gray-300 rounded-lg focus:outline-none focus:ring-2 focus:ring-blue-500">
49             </div>
50
51             <!-- Tombol Submit -->
52             <button type="submit"
53                 class="w-full bg-blue-600 text-white font-semibold py-2 rounded-lg hover:bg-blue-700 transition duration-200">
54                 Masuk
55             </button>
56         </form>
57     </div>
58
59 </body>
60 </html>
```

3. Membuat File Layout Utama (resources/views/layouts/app.blade.php)

Buat folder baru bernama layouts di dalam folder resources/views/, lalu buat file app.blade.php. File ini akan menjadi kerangka utama (Sidebar, Navbar, dan Konten Utama).

```
src > resources > views > layouts > app.blade.php
1 <!DOCTYPE html>
2 <html lang="id">
3 <head>
4 <meta charset="UTF-8">
5 <meta name="viewport" content="width=device-width, initial-scale=1.0">
6 <title>@yield('title', 'Dashboard Admin')</title>
7 <!-- Memuat Tailwind CSS CDN -->
8 <script src="https://cdn.tailwindcss.com"></script>
9 </head>
10 <body class="bg-gray-100 font-sans antialiased">
11
12 <div class="flex h-screen overflow-hidden">
13
14 <!-- SIDEBAR -->
15 <aside class="w-64 bg-gray-900 text-white flex flex-col hidden md:flex">
16 <div class="p-5 text-xl font-bold tracking-wider border-b border-gray-800">
17 PANEL ADMIN
18 </div>
19 <nav class="flex-1 p-4 space-y-2">
20 <a href="{{ route('admin.dashboard') }}" class="block px-4 py-2 rounded-lg bg-gray-800 text-white font-medium">Dashboard</a>
21 <a href="{{ route('admin.alat.index') }}" class="block px-4 py-2 rounded-lg text-gray-400 hover:bg-gray-800 hover:text-white transition">Kelola Alat</a>
22 <a href="{{ route('admin.user.index') }}" class="block px-4 py-2 rounded-lg text-gray-400 hover:bg-gray-800 hover:text-white transition">Kelola User</a>
23 </nav>
24 <div class="p-4 border-t border-gray-800 text-sm text-gray-400">
25 Logged in as: <span class="text-white font-semibold">{{ auth()->user()->name }}</span>
26 </div>
27 </aside>
28
29 <!-- MAIN CONTENT CONTAINER -->
30 <div class="flex-1 flex flex-col overflow-y-auto">
31
32 <!-- NAVBAR ATAS -->
33 <header class="bg-white shadow-sm h-16 flex items-center justify-between px-6 z-10">
34 <div class="text-lg font-semibold text-gray-800">
35 @yield('header-title', 'Dashboard')
36 </div>
37 <div>
38 <form action="{{ route('logout') }}" method="POST">
39 @csrf
40 <button type="submit" class="bg-red-500 hover:bg-red-600 text-white text-sm font-semibold px-4 py-2 rounded-lg transition">
41 Logout
42 </button>
43 </form>
44 </div>
45 </header>
46
47 <!-- KONTEN UTAMA HALAMAN -->
48 <main class="flex-1 p-6">
49 @yield('content')
50 </main>
51 </div>
52 </div>
53
54 </body>
55 </html>
```

4. Buat file baru bernama dashboard.blade.php di dalam folder resources/views/admin/.

```
1 @extends('layouts.app')
2
3 @section('title', 'Dashboard Admin - Sistem Peminjaman')
4 @section('header-title', 'Ringkasan Aktivitas Sistem')
5
6 @section('content')
7 <!-- Alert Selamat Datang -->
8 <div class="mb-6 bg-emerald-50 border border-emerald-200 text-emerald-800 p-4 rounded-lg shadow-sm">
9 Selamat datang, <strong class="font-semibold">{{ auth()->user()->name }}</strong>! Anda login sebagai hak akses
10 <span class="uppercase font-bold text-emerald-900">{{ auth()->user()->role }}</span>.
11 </div>
```

```

13     <!-- Tabel Log Aktivitas -->
14     <div class="bg-white rounded-lg shadow-sm overflow-hidden border border-gray-200">
15         <div class="p-5 border-b border-gray-200 bg-gray-50">
16             <h3 class="text-lg font-bold text-gray-800">Log Aktivitas Terbaru</h3>
17         </div>
18         <div class="overflow-x-auto">
19             <table class="w-full text-left border-collapse">
20                 <thead>
21                     <tr class="bg-gray-100 text-gray-600 text-sm uppercase tracking-wider">
22                         <th class="py-3 px-4 border-b">Waktu</th>
23                         <th class="py-3 px-4 border-b">User</th>
24                         <th class="py-3 px-4 border-b">Aktivitas</th>
25                     </tr>
26                 </thead>
27                 <tbody class="text-gray-700 text-sm">
28                     @forelse($logs as $log)
29                         <tr class="hover:bg-gray-50 transition">
30                             <td class="py-3 px-4 border-b">{{ $log->created_at }}</td>
31                             <td class="py-3 px-4 border-b font-medium text-gray-900">{{ $log->user->name ?? 'Sistem' }}</td>
32                             <td class="py-3 px-4 border-b">{{ $log->aktivitas }}</td>
33                         </tr>
34                     @empty
35                         <tr>
36                             <td colspan="3" class="py-4 text-center text-gray-500">Belum ada log aktivitas.</td>
37                         </tr>
38                     @endforelse
39                 </tbody>
40             </table>
41         </div>
42     </div>
43 @endsection

```

5. Halaman Dashboard / Katalog Peminjam

Buat file baru di resources/views/peminjam/katalog.blade.php (pastikan buat folder resources/views/peminjam/ terlebih dahulu).

Halaman ini menampilkan daftar alat yang tersedia serta form sederhana untuk mengajukan peminjaman.

```

src > resources > views > peminjam > katalog.blade.php
1  <!DOCTYPE html>
2  <html lang="id">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>Katalog Alat - Peminjam</title>
7      <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
8  </head>
9  <body class="bg-light">
10
11      <nav class="navbar navbar-expand-lg navbar-dark bg-primary mb-4">
12          <div class="container">
13              <a class="navbar-brand" href="#">Panel Peminjam</a>
14              <div class="d-flex">
15                  <a href="{{ route('peminjam.riwayat') }}" class="btn btn-outline-light btn-sm me-2">Riwayat Pinjam</a>
16                  <form action="{{ route('logout') }}" method="POST" class="d-inline">
17                      @csrf
18                      <button type="submit" class="btn btn-light btn-sm text-primary">Logout</button>
19                  </form>
20              </div>
21          </div>
22      </nav>

```

```

24     <div class="container">
25         @if(session('success'))
26             <div class="alert alert-success">{{ session('success') }}</div>
27         @endif
28         @if(session('error'))
29             <div class="alert alert-danger">{{ session('error') }}</div>
30         @endif
31
32         <h3 class="mb-3">Katalog Alat Tersedia</h3>
33
34         <form action="{{ route('peminjam.peminjaman.ajukan') }}" method="POST">
35             @csrf
36             <div class="card shadow-sm mb-4">
37                 <div class="card-body">
38                     <div class="mb-3">
39                         <label class="form-label">Rencana Tanggal Kembali</label>
40                         <input type="date" name="tgl_kembali_plan" class="form-control" required>
41                     </div>

```

```

43                 <table class="table table-bordered">
44                     <thead>
45                         <tr>
46                             <th width="50">Pilih</th>
47                             <th>Nama Alat</th>
48                             <th>Kategori</th>
49                             <th>Stok Tersedia</th>
50                             <th width="150">Jumlah Pinjam</th>
51                         </tr>
52                     </thead>
53                     <tbody>
54                         @forelse($alats as $index => $alat)
55                             <tr>
56                                 <td class="text-center">
57                                     <input type="checkbox" name="alat_id[]" value="{{ $alat->id }}" class="form-check-input">
58                                 </td>
59                                 <td>{{ $alat->nama_alat }}</td>
60                                 <td>{{ $alat->kategori->nama_kategori ?? '-' }}</td>
61                                 <td>{{ $alat->stok }}</td>
62                                 <td>
63                                     <input type="number" name="jumlah[]" class="form-control form-control-sm" value="1" min="1" max="{{ $alat->stok }}">
64                                 </td>
65                             </tr>
66                         @empty

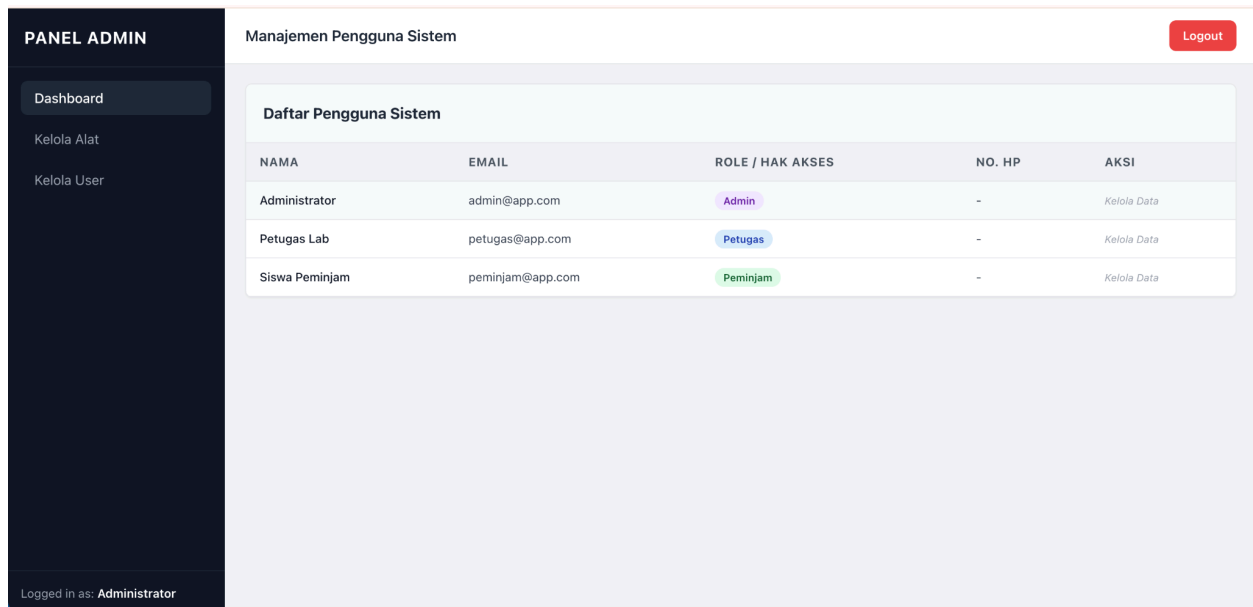
```

```

67                             <tr>
68                                 <td colspan="5" class="text-center">Tidak ada alat yang tersedia saat ini.</td>
69                             </tr>
70                         @endforelse
71                     </tbody>
72                 </table>
73                 <button type="submit" class="btn btn-primary">Ajukan Peminjaman</button>
74             </div>
75         </form>
76     </div>
77 </body>
78 </html>

```

MEMBUAT CRUD USER



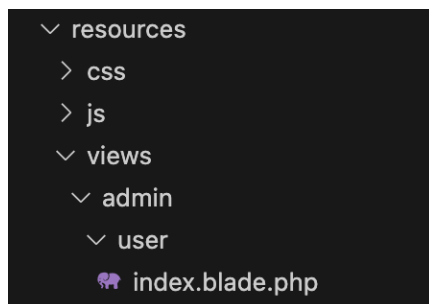
Langkah 1: Menambahkan Fungsi di AdminController.php

Buka controller admin kita, lalu pastikan method `indexUser` sudah menampilkan data pengguna dari database. Jika sebelumnya hanya memanggil view kosong, perbarui menjadi seperti ini:

```
48      // CRUD User (Manajemen User Admin, Petugas, Peminjam)
      1 reference | 0 overrides
49      public function indexUser()
50      {
51          $users = User::all();
52          return view('admin.user.index', compact('users'));
53      }
54  }
```

Langkah 2: Membuat File View Kelola User (resources/views/admin/user/index.blade.php)

Buat folder baru bernama `user` di dalam `resources/views/admin/`, lalu buat file baru bernama `index.blade.php`.



Kita akan menggunakan layout utama (`@extends('layouts.app')`) yang sama agar desain sidebar dan navbarnya konsisten:

```

1  @extends('layouts.app')
2
3  @section('title', 'Kelola User - Panel Admin')
4  @section('header-title', 'Manajemen Pengguna Sistem')
5
6  @section('content')
7      <!-- Notifikasi Sukses/Gagal -->
8      @if(session('success'))
9          <div class="mb-4 bg-emerald-50 border border-emerald-200 text-emerald-800 p-4 rounded-lg shadow-sm text-sm">
10              {{ session('success') }}
11          </div>
12      @endif
13
14      <div class="bg-white rounded-lg shadow-sm overflow-hidden border border-gray-200">
15          <div class="p-5 border-b border-gray-200 bg-gray-50 flex justify-between items-center">
16              <h3 class="text-lg font-bold text-gray-800">Daftar Pengguna Sistem</h3>
17              <!-- Tombol Tambah User (jika ingin dibuatkan form tambah) -->
18          </div>
19
20          <div class="overflow-x-auto">
21              <table class="w-full text-left border-collapse">
22                  <thead>
23                      <tr class="bg-gray-100 text-gray-600 text-sm uppercase tracking-wider">
24                          <th class="py-3 px-4 border-b">Nama</th>
25                          <th class="py-3 px-4 border-b">Email</th>
26                          <th class="py-3 px-4 border-b">Role / Hak Akses</th>
27                          <th class="py-3 px-4 border-b">No. HP</th>
28                          <th class="py-3 px-4 border-b">Aksi</th>
29                      </tr>
30                  </thead>

```

```

31              <tbody class="text-gray-700 text-sm">
32                  @forelse($users as $user)
33                      <tr class="hover:bg-gray-50 transition">
34                          <td class="py-3 px-4 border-b font-medium text-gray-900">{{ $user->name }}</td>
35                          <td class="py-3 px-4 border-b">{{ $user->email }}</td>
36                          <td class="py-3 px-4 border-b">
37                              <span class="px-2.5 py-1 text-xs font-semibold rounded-full
38                                  @if($user->role == 'admin') bg-purple-100 text-purple-800
39                                  @elseif($user->role == 'petugas') bg-blue-100 text-blue-800
40                                  @else bg-green-100 text-green-800 @endif">
41                                  {{ ucfirst($user->role) }}
42                              </span>
43                          </td>
44                          <td class="py-3 px-4 border-b">{{ $user->no_hp ?? '-' }}</td>
45                          <td class="py-3 px-4 border-b">
46                              <span class="text-gray-400 text-xs italic">Kelola Data</span>
47                          </td>
48                      </tr>
49                  @empty
50                      <tr>
51                          <td colspan="5" class="py-4 text-center text-gray-500">Belum ada data pengguna.</td>
52                      </tr>
53                  @endforelse
54              </tbody>
55          </table>
56      </div>
57  </div>
58 @endsection

```

MENAMBAHKAN TOMBOL AKSI DAN TAMBAH USER

Langkah 1: Update Tampilan Index User (resources/views/admin/user/index.blade.php)

Kita aktifkan tombol "Tambah User" di bagian atas dan menyematkan tombol Edit serta Hapus pada kolom aksi masing-masing baris tabel:

```
<div class="p-5 border-b border-gray-200 bg-gray-50 flex justify-between items-center">
  <h3 class="text-lg font-bold text-gray-800">Daftar Pengguna Sistem</h3>
  <a href="{{ route('admin.user.create') }}"
    class="bg-blue-600 hover:bg-blue-700 text-white text-sm font-semibold px-4 py-2 rounded-lg transition">
    + Tambah User
  </a>
</div>
```

Tambahkan juga tombol edit dan hapus di dalam setelah **<td class="py-3 px-4 border-b">**

```
<div class="flex items-center space-x-2">
  <!-- Tombol Edit -->
  <a href="{{ route('admin.user.edit', $user->id) }}"
    class="bg-amber-500 hover:bg-amber-600 text-white px-3 py-1.5 rounded text-xs font-semibold transition">
    Edit
  </a>

  <!-- Tombol Hapus -->
  <form action="{{ route('admin.user.destroy', $user->id) }}" method="POST" onsubmit="return confirm('Yakin ingin menghapus user ini?')">
    @csrf
    @method('DELETE')
    <button type="submit" class="bg-red-500 hover:bg-red-600 text-white px-3 py-1.5 rounded text-xs font-semibold transition">
      Hapus
    </button>
  </form>
</div>
```

Langkah 2: Membuat Form Tambah User (resources/views/admin/user/create.blade.php)

Buat file baru di folder resources/views/admin/user/ bernama create.blade.php:

```
1  @extends('layouts.app')
2
3  @section('title', 'Tambah User - Panel Admin')
4  @section('header-title', 'Tambah Pengguna Baru')
5
6  @section('content')
7  <div class="max-w-xl bg-white rounded-lg shadow-sm border-gray-200 p-6">
8    <form action="{{ route('admin.user.store') }}" method="POST">
9      @csrf
10
11      <div class="mb-4">
12        <label class="block text-gray-700 text-sm font-semibold mb-2">Nama Lengkap</label>
13        <input type="text" name="name" value="{{ old('name') }}" required
14          class="w-full px-3 py-2 border border-gray-300 rounded-lg focus:outline-none focus:ring-2 focus:ring-blue-500">
15        @error('name') <span class="text-red-500 text-xs">{{ $message }}</span> @enderror
16      </div>
17
18      <div class="mb-4">
19        <label class="block text-gray-700 text-sm font-semibold mb-2">Email</label>
20        <input type="email" name="email" value="{{ old('email') }}" required
21          class="w-full px-3 py-2 border border-gray-300 rounded-lg focus:outline-none focus:ring-2 focus:ring-blue-500">
22        @error('email') <span class="text-red-500 text-xs">{{ $message }}</span> @enderror
23      </div>
```

```

25     <div class="mb-4">
26         <label class="block text-gray-700 text-sm font-semibold mb-2">Password</label>
27         <input type="password" name="password" required
28             class="w-full px-3 py-2 border border-gray-300 rounded-lg focus:outline-none focus:ring-2 focus:ring-blue-500">
29         @error('password') <span class="text-red-500 text-xs">{{ $message }}</span> @enderror
30     </div>
31
32     <div class="mb-4">
33         <label class="block text-gray-700 text-sm font-semibold mb-2">Role / Hak Akses</label>
34         <select name="role" required class="w-full px-3 py-2 border border-gray-300 rounded-lg focus:outline-none focus:ring-2 focus:ring-blue-500">
35             <option value="peminjam">Peminjam</option>
36             <option value="petugas">Petugas</option>
37             <option value="admin">Admin</option>
38         </select>
39     </div>
40
41     <div class="mb-6">
42         <label class="block text-gray-700 text-sm font-semibold mb-2">No. HP (Opsional)</label>
43         <input type="text" name="no_hp" value="{{ old('no_hp') }}"
44             class="w-full px-3 py-2 border border-gray-300 rounded-lg focus:outline-none focus:ring-2 focus:ring-blue-500">
45     </div>
46
47     <div class="flex justify-end space-x-2">
48         <a href="{{ route('admin.user.index') }}"
49             class="bg-gray-300 hover:bg-gray-400 text-gray-800 px-4 py-2 rounded-lg text-sm font-semibold transition">Batal</a>
50         <button type="submit" class="bg-blue-600 hover:bg-blue-700 text-white px-4 py-2 rounded-lg text-sm font-semibold transition">Simpan</button>
51     </div>
52 </form>
53 </div>
54 @endsection

```

Langkah 3: Membuat Form Edit User (resources/views/admin/user/edit.blade.php)

Buat file baru di folder resources/views/admin/user/ bernama edit.blade.php:

```

src > resources > views > admin > user > edit.blade.php
1  @extends('layouts.app')
2
3  @section('title', 'Edit User - Panel Admin')
4  @section('header-title', 'Edit Data Pengguna')
5
6  @section('content')
7      <div class="max-w-xl bg-white rounded-lg shadow-sm border border-gray-200 p-6">
8          <form action="{{ route('admin.user.update', $user->id) }}" method="POST">
9              @csrf
10             @method('PUT')
11
12             <div class="mb-4">
13                 <label class="block text-gray-700 text-sm font-semibold mb-2">Nama Lengkap</label>
14                 <input type="text" name="name" value="{{ old('name', $user->name) }}" required
15                     class="w-full px-3 py-2 border border-gray-300 rounded-lg focus:outline-none focus:ring-2 focus:ring-blue-500">
16             </div>
17
18             <div class="mb-4">
19                 <label class="block text-gray-700 text-sm font-semibold mb-2">Email</label>
20                 <input type="email" name="email" value="{{ old('email', $user->email) }}" required
21                     class="w-full px-3 py-2 border border-gray-300 rounded-lg focus:outline-none focus:ring-2 focus:ring-blue-500">
22             </div>
23
24             <div class="mb-4">
25                 <label class="block text-gray-700 text-sm font-semibold mb-2">Password Baru
26                 <span class="text-xs text-gray-400 font-normal">(Kosongkan jika tidak ingin mengubah password)</span></label>
27                 <input type="password" name="password"
28                     class="w-full px-3 py-2 border border-gray-300 rounded-lg focus:outline-none focus:ring-2 focus:ring-blue-500">
29             </div>
30
31             <div class="mb-4">
32                 <label class="block text-gray-700 text-sm font-semibold mb-2">Role / Hak Akses</label>
33                 <select name="role" required class="w-full px-3 py-2 border border-gray-300 rounded-lg focus:outline-none focus:ring-2 focus:ring-blue-500">
34                     <option value="peminjam" {{ $user->role == 'peminjam' ? 'selected' : '' }}>Peminjam</option>
35                     <option value="petugas" {{ $user->role == 'petugas' ? 'selected' : '' }}>Petugas</option>
36                     <option value="admin" {{ $user->role == 'admin' ? 'selected' : '' }}>Admin</option>
37                 </select>
38             </div>
39

```

```

40     <div class="mb-6">
41         <label class="block text-gray-700 text-sm font-semibold mb-2">No. HP</label>
42         <input type="text" name="no_hp" value="{{ old('no_hp', $user->no_hp) }}"
43             class="w-full px-3 py-2 border border-gray-300 rounded-lg focus:outline-none focus:ring-2 focus:ring-blue-500">
44     </div>
45
46     <div class="flex justify-end space-x-2">
47         <a href="{{ route('admin.user.index') }}"
48             class="bg-gray-300 hover:bg-gray-400 text-gray-800 px-4 py-2 rounded-lg text-sm font-semibold transition">Batal</a>
49         <button type="submit"
50             class="bg-blue-600 hover:bg-blue-700 text-white px-4 py-2 rounded-lg text-sm font-semibold transition">Perbarui</button>
51     </div>
52 </form>
53 </div>
54 @endsection

```

Langkah 4: Pastikan Logika Controller Aktif (AdminController.php)

Pastikan AdminController memiliki fungsi create, store, edit, update, dan destroy untuk menangani alur di atas, tambahkan ini pada AdminController.php:

```

56     public function createUser()
57     {
58         return view('admin.user.create');
59     }
60
61     // Menyimpan user baru ke database
62     1 reference | 0 overrides
63     public function storeUser(Request $request)
64     {
65         $request->validate([
66             'name' => 'required|string|max:255',
67             'email' => 'required|string|email|max:255|unique:users',
68             'password' => 'required|string|min:6',
69             'role' => 'required|in:admin,petugas,peminjam',
70         ]);
71
72         User::create([
73             'name' => $request->name,
74             'email' => $request->email,
75             'password' => Hash::make($request->password),
76             'role' => $request->role,
77             'no_hp' => $request->no_hp,
78         ]);
79
80         return redirect()->route('admin.user.index')->with('success', 'User berhasil ditambahkan.');
```

```

82     public function editUser($id)
83     {
84         $user = User::findOrFail($id);
85         return view('admin.user.edit', compact('user'));
86     }
87
88     // Memperbarui data user
89     1 reference | 0 overrides
90     public function updateUser(Request $request, $id)
91     {
92         $user = User::findOrFail($id);
93
94         $request->validate([
95             'name' => 'required|string|max:255',
96             'email' => 'required|string|email|max:255|unique:users,email,' . $id,
97             'role' => 'required|in:admin,petugas,peminjam',
98         ]);

```

```

98     $data = [
99         'name' => $request->name,
100         'email' => $request->email,
101         'role' => $request->role,
102         'no_hp' => $request->no_hp,
103     ];
104
105     if ($request->filled('password')) {
106         $data['password'] = Hash::make($request->password);
107     }
108
109     $user->update($data);
110
111     return redirect()->route('admin.user.index')->with('success', 'Data user berhasil diperbarui.');
```

```

114     // Menghapus user
115     1 reference | 0 overrides
116     public function destroyUser($id)
117     {
118         $user = User::findOrFail($id);
119         $user->delete();
120
121         return redirect()->route('admin.user.index')->with('success', 'User berhasil dihapus.');
```

Langkah 5: Pastikan Route Terdaftar (routes/web.php)

Jika belum mendaftarkannya, pastikan routing untuk resource user ini sudah terpasang di dalam kelompok middleware admin (CRUD User):

```

//admin
Route::middleware(['auth', 'role:admin'])->prefix('admin')->name('admin.')->group(function () {
    Route::get('/dashboard', [AdminController::class, 'index'])->name('dashboard');

    // CRUD Alat
    Route::get('/alat', [AdminController::class, 'indexAlat'])->name('alat.index');
    Route::post('/alat', [AdminController::class, 'storeAlat'])->name('alat.store');

    // CRUD User
    Route::get('/users', [AdminController::class, 'indexUser'])->name('user.index');
    Route::get('/users/create', [AdminController::class, 'createUser'])->name('user.create');
    Route::post('/users', [AdminController::class, 'storeUser'])->name('user.store');
    Route::get('/users/{id}/edit', [AdminController::class, 'editUser'])->name('user.edit');
    Route::put('/users/{id}', [AdminController::class, 'updateUser'])->name('user.update');
    Route::delete('/users/{id}', [AdminController::class, 'destroyUser'])->name('user.destroy');
});
```

Tampilan Setelah jadi akan seperti ini

PANEL ADMIN

Dashboard

Kelola Alat

Kelola User

Manajemen Pengguna Sistem

Logout

User berhasil ditambahkan.

Daftar Pengguna Sistem

+ Tambah User

NAMA	EMAIL	ROLE / HAK AKSES	NO. HP	AKSI
Administrator	admin@app.com	Admin	-	Edit Hapus
Petugas Lab	petugas@app.com	Petugas	-	Edit Hapus
Fajar Oktavian	peminjam@app.com	Peminjam	-	Edit Hapus
Fadil	fadil@gmail.com	Petugas	-	Edit Hapus
ujang	ujang@gmail.com	Admin	-	Edit Hapus

PANEL ADMIN

Dashboard

Kelola Alat

Kelola User

Tambah Pengguna Baru

Logout

Nama Lengkap

Email

Password

Role / Hak Akses

Peminjam

No. HP (Opsional)

Batal

Simpan

PANEL ADMIN

Dashboard

Kelola Alat

Kelola User

Edit Data Pengguna

Logout

Nama Lengkap

ujang

Email

ujang@gmail.com

Password Baru (Kosongkan jika tidak ingin mengubah password)

Role / Hak Akses

Admin

No. HP

Batal

Perbarui

Tambahan fitur search dan pagination

Langkah 1: Perbarui Method indexUser di AdminController.php

Ubah method indexUser sebelumnya untuk menangkap input pencarian dari parameter request dan gunakan fungsi paginate() alih-alih get():

```
public function indexUser(Request $request)
{
    $search = $request->input('search');

    $users = User::when($search, function ($query, $search) {
        return $query->where('name', 'like', "%{$search}%")
            ->orWhere('email', 'like', "%{$search}%")
            ->orWhere('role', 'like', "%{$search}%");
    })
    ->latest()
    ->paginate(10) // Tampilkan 10 data per halaman
    ->withQueryString(); // Memastikan parameter search tetap ada saat pindah halaman

    return view('admin.user.index', compact('users', 'search'));
}
```

Langkah 2: Perbarui View resources/views/admin/user/index.blade.php

Tambahkan form input pencarian di bagian atas tabel dan komponen link paginasi ({{ \$users->links() }}) di bagian bawah tabel.

```
<h3 class="text-lg font-bold text-gray-800">Daftar Pengguna Sistem</h3>
<div class="flex items-center gap-3 w-full md:w-auto">
    <!-- Form Search -->
    <form action="{{ route('admin.user.index') }}" method="GET" class="flex w-full md:w-80">
        <input type="text" name="search" value="{{ request('search') }}" placeholder="Cari nama, email, role..."
            class="w-full px-3 py-2 text-sm border border-gray-300 rounded-l-lg focus:outline-none focus:ring-2 focus:ring-blue-500">
        <button type="submit" class="bg-gray-800 hover:bg-gray-900 text-white px-4 py-2 text-sm font-semibold rounded-r-lg transition">
            Cari
        </button>
        @if(request('search'))
            <a href="{{ route('admin.user.index') }}"
                class="ml-2 bg-gray-300 hover:bg-gray-400 text-gray-700 px-3 py-2 text-sm rounded-lg flex items-center transition" title="Reset Pencarian">
                Reset
            </a>
        @endif
    </form>

    <!-- Tombol Tambah User -->
    <a href="{{ route('admin.user.create') }}"
        class="bg-blue-600 hover:bg-blue-700 text-white text-sm font-semibold px-4 py-2 rounded-lg transition whitespace-nowrap">
        + Tambah User
    </a>
</div>
```

Simpan code dibawah ini setelah </table> </div>

```
90         </tbody>
91     </table>
92 </div>
93
94     <div class="p-4 border-t border-gray-200 bg-gray-50">
95         {{ $users->links() }}
96     </div>
97 </div>
98 @endsection
```

Catatan Penting untuk Styling Tailwind & Pagination Laravel 11/12:

Secara default, Laravel menggunakan styling pagination bawaan Tailwind. Jika tampilannya ingin terlihat lebih rapi dan menyatu dengan tema kustom, kita bisa mengeksekusi perintah artisan berikut di terminal untuk mempublikasikan file view pagination Laravel:

docker exec -it alat_app php artisan vendor:publish --tag=laravel-pagination

```
fajaroktaviannugraha@MacBook-Air-Fajar project-peminjaman-alat % docker exec -it alat_app php artisan vendor:publish --tag=laravel-pagination
INFO Publishing [laravel-pagination] assets.
Copying directory [vendor/laravel/framework/src/Illuminate/Pagination/resources/views] to [resources/views/vendor/pagination] ..... DONE
```

MEMBUAT CRUD KATEGORI

Langkah 1: Tambahkan Method di AdminController.php

Tambahkan fungsi-fungsi berikut ke dalam AdminController untuk mengatur logika CRUD kategori:

```
133     public function indexKategori(Request $request)
134     {
135         $search = $request->input('search');
136
137         $kategoris = Kategori::when($search, function ($query, $search) {
138             return $query->where('nama_kategori', 'like', "%{$search}%");
139         })
140             ->latest()
141             ->paginate(5)
142             ->withQueryString();
143
144         return view('admin.kategori.index', compact('kategoris', 'search'));
145     }
146
147     // 2. Menampilkan form tambah kategori
148     1 reference | 0 overrides
149     public function createKategori()
150     {
151         return view('admin.kategori.create');
152     }
153
154     // 3. Menyimpan kategori baru
155     1 reference | 0 overrides
156     public function storeKategori(Request $request)
157     {
158         $request->validate([
159             'nama_kategori' => 'required|string|max:255|unique:kategoris,nama_kategori',
160         ]);
161
162         Kategori::create([
163             'nama_kategori' => $request->nama_kategori,
164         ]);
165
166         return redirect()->route('admin.kategori.index')->with('success', 'Kategori berhasil ditambahkan.');
```

```
167     // 4. Menampilkan form edit kategori
168     1 reference | 0 overrides
169     public function editKategori($id)
170     {
171         $kategori = Kategori::findOrFail($id);
172         return view('admin.kategori.edit', compact('kategori'));
173     }
174
175     // 5. Memperbarui kategori
176     1 reference | 0 overrides
177     public function updateKategori(Request $request, $id)
178     {
179         $kategori = Kategori::findOrFail($id);
180
181         $request->validate([
182             'nama_kategori' => 'required|string|max:255|unique:kategoris,nama_kategori,' . $id,
183         ]);
184
185         $kategori->update([
186             'nama_kategori' => $request->nama_kategori,
187         ]);
188
189         return redirect()->route('admin.kategori.index')->with('success', 'Kategori berhasil diperbarui.');
```

```

189 // 6. Menghapus kategori
190 1 reference | 0 overrides
191 public function destroyKategori($id)
192 {
193     $kategori = Kategori::findOrFail($id);
194
195     // Optional: Cek apakah kategori masih dipakai oleh alat
196     if ($kategori->alats()->count() > 0) {
197         return redirect()->route('admin.kategori.index')
198             ->with('error', 'Kategori tidak dapat dihapus karena masih digunakan oleh data alat.');
199     }
200
201     $kategori->delete();
202
203     return redirect()->route('admin.kategori.index')->with('success', 'Kategori berhasil dihapus.');
```

Langkah 2: Daftarkan Route di routes/web.php

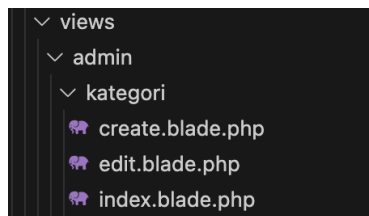
Tambahkan baris route berikut di dalam kelompok middleware admin:

```

29 // CRUD Kategori
30 Route::get('/kategori', [AdminController::class, 'indexKategori'])->name('kategori.index');
31 Route::get('/kategori/create', [AdminController::class, 'createKategori'])->name('kategori.create');
32 Route::post('/kategori', [AdminController::class, 'storeKategori'])->name('kategori.store');
33 Route::get('/kategori/{id}/edit', [AdminController::class, 'editKategori'])->name('kategori.edit');
34 Route::put('/kategori/{id}', [AdminController::class, 'updateKategori'])->name('kategori.update');
35 Route::delete('/kategori/{id}', [AdminController::class, 'destroyKategori'])->name('kategori.destroy');
```

Langkah 3: Buat Tampilan View

Buat folder baru bernama kategori di dalam resources/views/admin/, lalu buat 3 file view berikut:



A. resources/views/admin/kategori/index.blade.php

```

src > resources > views > admin > kategori > index.blade.php
1  @extends('layouts.app')
2
3  @section('title', 'Kelola Kategori - Panel Admin')
4  @section('header-title', 'Manajemen Kategori Alat')
5
6  @section('content')
7      <!-- Notifikasi -->
8      @if(session('success'))
9          <div class="mb-4 bg-emerald-50 border border-emerald-200 text-emerald-800 p-4 rounded-lg shadow-sm text-sm">
10              {{ session('success') }}
11          </div>
12      @endif
13      @if(session('error'))
14          <div class="mb-4 bg-red-50 border border-red-200 text-red-800 p-4 rounded-lg shadow-sm text-sm">
15              {{ session('error') }}
16          </div>
17      @endif
18
19      <div class="bg-white rounded-lg shadow-sm overflow-hidden border border-gray-200">
20          <div class="p-5 border-b border-gray-200 bg-gray-50 flex flex-col md:flex-row justify-between items-center gap-4">
21              <h3 class="text-lg font-bold text-gray-800">Daftar Kategori Alat</h3>
22          </div>
```

```

23         <div class="flex items-center gap-3 w-full md:w-auto">
24             <!-- Form Search -->
25             <form action="{{ route('admin.kategori.index') }}" method="GET" class="flex w-full md:w-80">
26                 <input type="text" name="search" value="{{ request('search') }}" placeholder="Cari nama kategori..."
27                     class="w-full px-3 py-2 text-sm border border-gray-300 rounded-l-lg focus:outline-none focus:ring-2 focus:ring-blue-500">
28                 <button type="submit" class="bg-gray-800 hover:bg-gray-900 text-white px-4 py-2 text-sm font-semibold rounded-r-lg transition">
29                     Cari
30                 </button>
31                 @if(request('search'))
32                     <a href="{{ route('admin.kategori.index') }}"
33                         class="ml-2 bg-gray-300 hover:bg-gray-400 text-gray-700 px-3 py-2 text-sm rounded-lg flex items-center transition">
34                         Reset
35                     </a>
36                 @endif
37             </form>
38
39             <!-- Tombol Tambah -->
40             <a href="{{ route('admin.kategori.create') }}"
41                 class="bg-blue-600 hover:bg-blue-700 text-white text-sm font-semibold px-4 py-2 rounded-lg transition whitespace-nowrap">
42                 + Tambah Kategori
43             </a>
44         </div>
45     </div>
46
47     <div class="overflow-x-auto">
48         <table class="w-full text-left border-collapse">
49             <thead>
50                 <tr class="bg-gray-100 text-gray-600 text-sm uppercase tracking-wider">
51                     <th class="py-3 px-4 border-b w-16 text-center">No</th>
52                     <th class="py-3 px-4 border-b">Nama Kategori</th>
53                     <th class="py-3 px-4 border-b w-48">Aksi</th>
54                 </tr>
55             </thead>
56             <tbody class="text-gray-700 text-sm">
57                 @forelse($kategoris as $index => $kategori)
58                     <tr class="hover:bg-gray-50 transition">
59                         <td class="py-3 px-4 border-b text-center">{{ $kategoris->firstItem() + $index }}</td>
60                         <td class="py-3 px-4 border-b font-medium text-gray-900">{{ $kategori->nama_kategori }}</td>
61                         <td class="py-3 px-4 border-b">
62                             <div class="flex items-center space-x-2">
63                                 <a href="{{ route('admin.kategori.edit', $kategori->id) }}"
64                                     class="bg-amber-500 hover:bg-amber-600 text-white px-3 py-1.5 rounded text-xs font-semibold transition">
65                                     Edit
66                                 </a>
67
68                                 <form action="{{ route('admin.kategori.destroy', $kategori->id) }}"
69                                     method="POST" onsubmit="return confirm('Yakin ingin menghapus kategori ini?')">
70                                     @csrf
71                                     @method('DELETE')
72                                     <button type="submit"
73                                         class="bg-red-500 hover:bg-red-600 text-white px-3 py-1.5 rounded text-xs font-semibold transition">
74                                         Hapus
75                                     </button>
76                                 </form>
77                             </div>
78                         </td>
79                     </tr>
80                     @empty
81                         <tr>
82                             <td colspan="3" class="py-4 text-center text-gray-500">Belum ada data kategori.</td>
83                         </tr>
84                     @endforelse
85                 </tbody>
86             </table>
87         </div>
88
89         <!-- Pagination -->
90         <div class="p-4 border-t border-gray-200 bg-gray-50">
91             {{ $kategoris->links() }}
92         </div>
93     </div>
94 @endsection

```

B. resources/views/admin/kategori/create.blade.php

```

src > resources > views > admin > kategori > create.blade.php
1  @extends('layouts.app')
2
3  @section('title', 'Tambah Kategori - Panel Admin')
4  @section('header-title', 'Tambah Kategori Baru')
5
6  @section('content')
7  <div class="max-w-xl bg-white rounded-lg shadow-sm border border-gray-200 p-6">
8      <form action="{{ route('admin.kategori.store') }}" method="POST">
9          @csrf
10
11          <div class="mb-4">
12              <label class="block text-gray-700 text-sm font-semibold mb-2">Nama Kategori</label>
13              <input type="text" name="nama_kategori" value="{{ old('nama_kategori') }}" required placeholder="Contoh: Jaringan, Mikrokontroler, Power Tools"
14                  class="w-full px-3 py-2 border border-gray-300 rounded-lg focus:outline-none focus:ring-2 focus:ring-blue-500">
15              @error('nama_kategori') <span class="text-red-500 text-xs">{{ $message }}</span> @enderror
16          </div>
17
18          <div class="flex justify-end space-x-2">
19              <a href="{{ route('admin.kategori.index') }}"
20                  class="bg-gray-300 hover:bg-gray-400 text-gray-800 px-4 py-2 rounded-lg text-sm font-semibold transition">Batal</a>
21              <button type="submit" class="bg-blue-600 hover:bg-blue-700 text-white px-4 py-2 rounded-lg text-sm font-semibold transition">Simpan</button>
22          </div>
23      </form>
24  </div>
25  @endsection

```

C. resources/views/admin/kategori/edit.blade.php

```

src > resources > views > admin > kategori > edit.blade.php
1  @extends('layouts.app')
2
3  @section('title', 'Edit Kategori - Panel Admin')
4  @section('header-title', 'Edit Kategori Alat')
5
6  @section('content')
7  <div class="max-w-xl bg-white rounded-lg shadow-sm border border-gray-200 p-6">
8      <form action="{{ route('admin.kategori.update', $kategori->id) }}" method="POST">
9          @csrf
10          @method('PUT')
11
12          <div class="mb-4">
13              <label class="block text-gray-700 text-sm font-semibold mb-2">Nama Kategori</label>
14              <input type="text" name="nama_kategori" value="{{ old('nama_kategori', $kategori->nama_kategori) }}" required
15                  class="w-full px-3 py-2 border border-gray-300 rounded-lg focus:outline-none focus:ring-2 focus:ring-blue-500">
16              @error('nama_kategori') <span class="text-red-500 text-xs">{{ $message }}</span> @enderror
17          </div>
18
19          <div class="flex justify-end space-x-2">
20              <a href="{{ route('admin.kategori.index') }}"
21                  class="bg-gray-300 hover:bg-gray-400 text-gray-800 px-4 py-2 rounded-lg text-sm font-semibold transition">Batal</a>
22              <button type="submit"
23                  class="bg-blue-600 hover:bg-blue-700 text-white px-4 py-2 rounded-lg text-sm font-semibold transition">Perbarui</button>
24          </div>
25      </form>
26  </div>
27  @endsection

```

jangan lupa juga untuk menambahkan menu Kelola Kategori pada bagian Sidebar di file layout utama (resources/views/layouts/app.blade.php) agar mudah diakses:

```

<nav class="flex-1 p-4 space-y-2">
    <a href="{{ route('admin.dashboard') }}"
    class="block px-4 py-2 rounded-lg bg-gray-800 text-white font-medium">Dashboard</a>
    <a href="{{ route('admin.user.index') }}"
    class="block px-4 py-2 rounded-lg text-gray-400 hover:bg-gray-800 hover:text-white transition">Kelola User</a>
    <a href="{{ route('admin.kategori.index') }}"
    class="block px-4 py-2 rounded-lg text-gray-400 hover:bg-gray-800 hover:text-white transition">Kelola Kategori</a>
    <a href="{{ route('admin.alat.index') }}"
    class="block px-4 py-2 rounded-lg text-gray-400 hover:bg-gray-800 hover:text-white transition">Kelola Alat</a>
</nav>

```

Tampilan halaman kategori

PANEL ADMIN

Dashboard

Kelola User

Kelola Kategori

Kelola Alat

Manajemen Kategori Alat

Logout

Daftar Kategori Alat

Cari nama kategori...

Cari

+ Tambah Kategori

NO	NAMA KATEGORI	AKSI
1	Furniture	Edit Hapus
2	Alat Pertukangan	Edit Hapus
3	Alat Kebersihan	Edit Hapus
4	Alat Perkakas	Edit Hapus
5	Alat Elektronik	Edit Hapus

Showing 1 to 5 of 6 results

<

1

2

>

PANEL ADMIN

Dashboard

Kelola User

Kelola Kategori

Kelola Alat

Tambah Kategori Baru

Nama Kategori

Contoh: Jaringan, Mikrokontroler, Power Tools

Batal

Simpan

PANEL ADMIN

Dashboard

Kelola User

Kelola Kategori

Kelola Alat

Edit Kategori Alat

Nama Kategori

Alat Pertukangan

Batal

Perbarui

MEMBUAT CRUD ALAT

1. Perbarui Method Alat di AdminController.php

Sesuaikan fungsi indexAlat, storeAlat, dan updateAlat di controller menjadi seperti ini:

```
21 // CRUD Alat: Menampilkan daftar alat
    1 reference | 0 overrides
22 public function indexAlat(Request $request)
23 {
24     $search = $request->input('search');
25
26     $alats = Alat::with('kategori')
27         ->when($search, function ($query, $search) {
28             return $query->where('nama_alat', 'like', "%{$search}%")
29                 ->orWhere('status_kondisi', 'like', "%{$search}%")
30                 ->orWhereHas('kategori', function ($q) use ($search) {
31                     $q->where('nama_kategori', 'like', "%{$search}%");
32                 });
33         })
34         ->latest()
35         ->paginate(10)
36         ->withQueryString();
37
38     return view('admin.alat.index', compact('alats', 'search'));
39 }
```

```
41 // 2. Menampilkan form tambah alat
    1 reference | 0 overrides
42 public function createAlat()
43 {
44     $kategoris = Kategori::all();
45     return view('admin.alat.create', compact('kategoris'));
46 }
47
48 // 3. Menyimpan alat baru
    1 reference | 0 overrides
49 public function storeAlat(Request $request)
50 {
51     $request->validate([
52         'nama_alat' => 'required|string|max:255',
53         'kategori_id' => 'required|exists:kategoris,id',
54         'stok' => 'required|integer|min:0',
55         'status_kondisi' => 'required|string|max:100',
56         'deskripsi' => 'nullable|string',
57         'gambar' => 'nullable|image|mimes:jpeg,png,jpg|max:2048',
58     ]);
59
60     $data = $request->all();
61
62     // Handle Upload Gambar jika ada
63     if ($request->hasFile('gambar')) {
64         $file = $request->file('gambar');
65         $filename = time() . '_' . $file->getClientOriginalName();
66         $file->move(public_path('storage/alat'), $filename);
67         $data['gambar'] = 'storage/alat/' . $filename;
68     }
69 }
```

```

70     Alat::create($data);
71
72     return redirect()->route('admin.alat.index')->with('success', 'Data alat berhasil ditambahkan.');
```

```

75     // 4. Menampilkan form edit alat
76     1 reference | 0 overrides
77     public function editAlat($id)
78     {
79         $alat = Alat::findOrFail($id);
80         $kategoris = Kategori::all();
81         return view('admin.alat.edit', compact('alat', 'kategoris'));

```

```

83     // 5. Memperbarui data alat
84     1 reference | 0 overrides
85     public function updateAlat(Request $request, $id)
86     {
87         $alat = Alat::findOrFail($id);
88
89         $request->validate([
90             'nama_alat' => 'required|string|max:255',
91             'kategori_id' => 'required|exists:kategoris,id',
92             'stok' => 'required|integer|min:0',
93             'status_kondisi' => 'required|string|max:100',
94             'deskripsi' => 'nullable|string',
95             'gambar' => 'nullable|image|mimes:jpeg,png,jpg|max:2048',
96         ]);
97
98         $data = $request->all();
99
100        // Handle Update Gambar jika ada file baru
101        if ($request->hasFile('gambar')) {
102            // Hapus gambar lama jika ada
103            if ($alat->gambar && file_exists(public_path($alat->gambar))) {
104                unlink(public_path($alat->gambar));
105            }
106
107            $file = $request->file('gambar');
108            $filename = time() . '_' . $file->getClientOriginalName();
109            $file->move(public_path('storage/alat'), $filename);
110            $data['gambar'] = 'storage/alat/' . $filename;
111        }
112
113        $alat->update($data);
114
115        return redirect()->route('admin.alat.index')->with('success', 'Data alat berhasil diperbarui.');
```

```

117    // 6. Menghapus data alat
118    1 reference | 0 overrides
119    public function destroyAlat($id)
120    {
121        $alat = Alat::findOrFail($id);
122
123        // Hapus file gambar fisik jika ada
124        if ($alat->gambar && file_exists(public_path($alat->gambar))) {
125            unlink(public_path($alat->gambar));
126        }
127
128        $alat->delete();
129
130        return redirect()->route('admin.alat.index')->with('success', 'Data alat berhasil dihapus.');
```

2. Buat Tampilan Views Kelola Alat

Buat folder baru bernama alat di dalam resources/views/admin/, lalu buat 3 file berikut:

A. resources/views/admin/alat/index.blade.php

```
1 @extends('layouts.app')
2
3 @section('title', 'Kelola Alat - Panel Admin')
4 @section('header-title', 'Manajemen Data Alat')
5
6 @section('content')
7     @if(session('success'))
8         <div class="mb-4 bg-emerald-50 border border-emerald-200 text-emerald-800 p-4 rounded-lg shadow-sm text-sm">
9             {{ session('success') }}
10        </div>
11    @endif
12
13    <div class="bg-white rounded-lg shadow-sm overflow-hidden border border-gray-200">
14        <div class="p-5 border-b border-gray-200 bg-gray-50 flex flex-col md:flex-row justify-between items-center gap-4">
15            <h3 class="text-lg font-bold text-gray-800">Daftar Alat Laboratorium</h3>
16
17            <div class="flex items-center gap-3 w-full md:w-auto">
18                <!-- Form Search -->
19                <form action="{{ route('admin.alat.index') }}" method="GET" class="flex w-full md:w-80">
20                    <input type="text" name="search" value="{{ request('search') }}" placeholder="Cari nama alat, kategori..."
21                        class="w-full px-3 py-2 text-sm border border-gray-300 rounded-l-lg focus:outline-none focus:ring-2 focus:ring-blue-500">
22                    <button type="submit" class="bg-gray-800 hover:bg-gray-900 text-white px-4 py-2 text-sm font-semibold rounded-r-lg transition">
23                        Cari
24                    </button>
25                    @if(request('search'))
26                        <a href="{{ route('admin.alat.index') }}"
27                            class="ml-2 bg-gray-300 hover:bg-gray-400 text-gray-700 px-3 py-2 text-sm rounded-lg flex items-center transition">
28                            Reset
29                        </a>
30                    @endif
31                </form>
32
33                <!-- Tombol Tambah -->
34                <a href="{{ route('admin.alat.create') }}"
35                    class="bg-blue-600 hover:bg-blue-700 text-white text-sm font-semibold px-4 py-2 rounded-lg transition whitespace-nowrap">
36                    + Tambah Alat
37                </a>
38            </div>
39        </div>
40    </div>
```

```

41         <div class="overflow-x-auto">
42             <table class="w-full text-left border-collapse">
43                 <thead>
44                     <tr class="bg-gray-100 text-gray-600 text-sm uppercase tracking-wider">
45                         <th class="py-3 px-4 border-b">Gambar</th>
46                         <th class="py-3 px-4 border-b">Nama Alat</th>
47                         <th class="py-3 px-4 border-b">Kategori</th>
48                         <th class="py-3 px-4 border-b">Stok</th>
49                         <th class="py-3 px-4 border-b">Kondisi</th>
50                         <th class="py-3 px-4 border-b">Aksi</th>
51                     </tr>
52                 </thead>
53                 <tbody class="text-gray-700 text-sm">
54                     @forelse($alats as $alat)
55                         <tr class="hover:bg-gray-50 transition">
56                             <td class="py-3 px-4 border-b">
57                                 @if($alat->gambar)
58                                     nama_alat }}" class="w-12 h-12 object-cover rounded-lg border">
59                                 @else
60                                     <span class="text-xs text-gray-400 italic">Tidak ada</span>
61                                 @endif
62                             </td>
63                             <td class="py-3 px-4 border-b font-medium text-gray-900">{{ $alat->nama_alat }}</td>
64                             <td class="py-3 px-4 border-b">{{ $alat->kategori->nama_kategori ?? '-' }}</td>
65                             <td class="py-3 px-4 border-b font-semibold">{{ $alat->stok }}</td>
66                             <td class="py-3 px-4 border-b">
67                                 <span class="px-2.5 py-1 text-xs font-semibold rounded-full
68                                     @if(strtolower($alat->status_kondisi) == 'baik') bg-emerald-100 text-emerald-800
69                                     @else bg-amber-100 text-amber-800 @endif">
70                                     {{ $alat->status_kondisi }}
71                                 </span>
72                             </td>
73                             <td class="py-3 px-4 border-b">
74                                 <div class="flex items-center space-x-2">
75                                     <a href="{{ route('admin.alat.edit', $alat->id) }}"
76                                     class="bg-amber-500 hover:bg-amber-600 text-white px-3 py-1.5 rounded text-xs font-semibold transition">
77                                         Edit
78                                 </div>
79                             </td>
80                             <td>
81                                 <form action="{{ route('admin.alat.destroy', $alat->id) }}"
82                                 method="POST" onsubmit="return confirm('Yakin ingin menghapus alat ini?')">
83                                     @csrf
84                                     @method('DELETE')
85                                     <button type="submit" class="bg-red-500 hover:bg-red-600 text-white px-3 py-1.5 rounded text-xs font-semibold transition">
86                                         Hapus
87                                     </button>
88                                 </form>
89                             </td>
90                         </tr>
91                     @empty
92                         <tr>
93                             <td colspan="6" class="py-4 text-center text-gray-500">Belum ada data alat.</td>
94                         </tr>
95                     @endforelse
96                 </tbody>
97             </table>
98         </div>
99
100         <div class="p-4 border-t border-gray-200 bg-gray-50">
101             {{ $alats->links() }}
102         </div>
103     </div>
104 @endsection

```

B. resources/views/admin/alat/create.blade.php

```

src > resources > views > admin > alat > create.blade.php
1  @extends('layouts.app')
2
3  @section('title', 'Tambah Alat - Panel Admin')
4  @section('header-title', 'Tambah Alat Baru')
5
6  @section('content')
7      <div class="max-w-2xl bg-white rounded-lg shadow-sm border border-gray-200 p-6">
8          <form action="{{ route('admin.alat.store') }}" method="POST" enctype="multipart/form-data">
9              @csrf
10

```

```

11     <div class="mb-4">
12         <label class="block text-gray-700 text-sm font-semibold mb-2">Nama Alat</label>
13         <input type="text" name="nama_alat" value="{{ old('nama_alat') }}" required placeholder="Contoh: Multimeter Digital"
14             class="w-full px-3 py-2 border border-gray-300 rounded-lg focus:outline-none focus:ring-2 focus:ring-blue-500">
15         @error('nama_alat') <span class="text-red-500 text-xs">{{ $message }}</span> @enderror
16     </div>
17
18     <div class="mb-4">
19         <label class="block text-gray-700 text-sm font-semibold mb-2">Kategori</label>
20         <select name="kategori_id" required class="w-full px-3 py-2 border border-gray-300 rounded-lg focus:outline-none focus:ring-2 focus:ring-blue-500">
21             <option value="">-- Pilih Kategori --</option>
22             @foreach($kategoris as $kategori)
23                 <option value="{{ $kategori->id }}" {{ old('kategori_id') == $kategori->id ? 'selected' : '' }}>
24                     {{ $kategori->nama_kategori }}
25                 </option>
26             @endforeach
27         </select>
28         @error('kategori_id') <span class="text-red-500 text-xs">{{ $message }}</span> @enderror
29     </div>
30
31     <div class="grid grid-cols-1 md:grid-cols-2 gap-4 mb-4">
32         <div>
33             <label class="block text-gray-700 text-sm font-semibold mb-2">Stok</label>
34             <input type="number" name="stok" value="{{ old('stok', 1) }}" min="0" required
35                 class="w-full px-3 py-2 border border-gray-300 rounded-lg focus:outline-none focus:ring-2 focus:ring-blue-500">
36             @error('stok') <span class="text-red-500 text-xs">{{ $message }}</span> @enderror
37         </div>
38     </div>
39
40     <div class="block text-gray-700 text-sm font-semibold mb-2">Status Kondisi</label>
41     <input type="text" name="status_kondisi" value="{{ old('status_kondisi', 'Baik') }}" required placeholder="Contoh: Baik / Rusak Ringan"
42         class="w-full px-3 py-2 border border-gray-300 rounded-lg focus:outline-none focus:ring-2 focus:ring-blue-500">
43     @error('status_kondisi') <span class="text-red-500 text-xs">{{ $message }}</span> @enderror
44 </div>
45
46 <div class="mb-4">
47     <label class="block text-gray-700 text-sm font-semibold mb-2">Deskripsi (Optional)</label>
48     <textarea name="rows=3" name="deskripsi" placeholder="Keterangan tambahan tentang alat..."
49         class="w-full px-3 py-2 border border-gray-300 rounded-lg focus:outline-none focus:ring-2 focus:ring-blue-500">{{ old('deskripsi') }}</textarea>
50     @error('deskripsi') <span class="text-red-500 text-xs">{{ $message }}</span> @enderror
51 </div>
52
53 <div class="mb-6">
54     <label class="block text-gray-700 text-sm font-semibold mb-2">Gambar Alat (Optional)</label>
55     <input type="file" name="gambar" accept="image/*"
56         class="w-full text-sm text-gray-500 file:mr-4 file:py-2 file:px-4 file:rounded-lg file:border-0
57             file:text-sm file:font-semibold file:bg-blue-50 file:text-blue-700 hover:file:bg-blue-100">
58     @error('gambar') <span class="text-red-500 text-xs">{{ $message }}</span> @enderror
59 </div>
60
61 <div class="flex justify-end space-x-2">
62     <a href="{{ route('admin.alat.index') }}"
63         class="bg-gray-300 hover:bg-gray-400 text-gray-800 px-4 py-2 rounded-lg text-sm font-semibold transition">Batal</a>
64     <button type="submit"
65         class="bg-blue-600 hover:bg-blue-700 text-white px-4 py-2 rounded-lg text-sm font-semibold transition">Simpan</button>
66 </div>
67 </form>
68 </div>
69 @endsection

```

C. resources/views/admin/alat/edit.blade.php

```

src > resources > views > admin > alat > edit.blade.php
1  @extends('layouts.app')
2
3  @section('title', 'Edit Alat - Panel Admin')
4  @section('header-title', 'Edit Data Alat')
5
6  @section('content')
7  <div class="max-w-2xl bg-white rounded-lg shadow-sm border border-gray-200 p-6">
8      <form action="{{ route('admin.alat.update', $alat->id) }}" method="POST" enctype="multipart/form-data">
9          @csrf
10         @method('PUT')
11
12         <div class="mb-4">
13             <label class="block text-gray-700 text-sm font-semibold mb-2">Nama Alat</label>
14             <input type="text" name="nama_alat" value="{{ old('nama_alat', $alat->nama_alat) }}" required
15                 class="w-full px-3 py-2 border border-gray-300 rounded-lg focus:outline-none focus:ring-2 focus:ring-blue-500">
16         </div>
17
18         <div class="mb-4">
19             <label class="block text-gray-700 text-sm font-semibold mb-2">Kategori</label>
20             <select name="kategori_id" required class="w-full px-3 py-2 border border-gray-300 rounded-lg focus:outline-none focus:ring-2 focus:ring-blue-500">
21                 @foreach($kategoris as $kategori)

```

```

22         <option value="{{ $kategori->id }}" {{ old('kategori_id', $alat->kategori_id) == $kategori->id ? 'selected' : '' }}>
23             {{ $kategori->nama_kategori }}
24         </option>
25     @endforeach
26 </select>
27 </div>
28
29 <div class="grid grid-cols-1 md:grid-cols-2 gap-4 mb-4">
30     <div>
31         <label class="block text-gray-700 text-sm font-semibold mb-2">Stok</label>
32         <input type="number" name="stok" value="{{ old('stok', $alat->stok) }}" min="0" required
33             class="w-full px-3 py-2 border border-gray-300 rounded-lg focus:outline-none focus:ring-2 focus:ring-blue-500">
34     </div>
35     <div>
36         <label class="block text-gray-700 text-sm font-semibold mb-2">Status Kondisi</label>
37         <input type="text" name="status_kondisi" value="{{ old('status_kondisi', $alat->status_kondisi) }}" required
38             class="w-full px-3 py-2 border border-gray-300 rounded-lg focus:outline-none focus:ring-2 focus:ring-blue-500">
39     </div>
40 </div>
41
42 <div class="mb-4">
43     <label class="block text-gray-700 text-sm font-semibold mb-2">Deskripsi</label>
44     <textarea name="deskripsi" rows="3"
45         class="w-full px-3 py-2 border border-gray-300 rounded-lg focus:outline-none focus:ring-2 focus:ring-blue-500">
46         {{ old('deskripsi', $alat->deskripsi) }}</textarea>
47 </div>
48
49 <div class="mb-6">
50     <label class="block text-gray-700 text-sm font-semibold mb-2">Gambar Alat <span class="text-xs text-gray-400 font-normal">
51         (Biarkan kosong jika tidak ingin mengubah gambar)</span></label>
52     @if($alat->gambar)
53         <div class="mb-2">
54             
55         </div>
56     @endif
57     <input type="file" name="gambar" accept="image/*"
58         class="w-full text-sm text-gray-500 file:mr-4 file:py-2 file:px-4 file:rounded-lg file:border-0
59         file:text-sm file:font-semibold file:bg-blue-50 file:text-blue-700 hover:file:bg-blue-100">
60 </div>
61
62 <div class="flex justify-end space-x-2">
63     <a href="{{ route('admin.alat.index') }}"
64         class="bg-gray-300 hover:bg-gray-400 text-gray-800 px-4 py-2 rounded-lg text-sm font-semibold transition">Batal</a>
65     <button type="submit" class="bg-blue-600 hover:bg-blue-700 text-white px-4 py-2 rounded-lg text-sm font-semibold transition">Perbarui</button>
66 </div>
67 </form>
68 </div>
69 @endsection

```

3. Daftarkan Route Alat di routes/web.php

Pastikan rute resource alat sudah aktif di dalam grup middleware admin:

```

//admin
Route::middleware(['auth', 'role:admin'])->prefix('admin')->name('admin.')->group(function () {
    Route::get('/dashboard', [AdminController::class, 'index'])->name('dashboard');

    // CRUD Alat
    Route::get('/alat', [AdminController::class, 'indexAlat'])->name('alat.index');
    Route::get('/alat/create', [AdminController::class, 'createAlat'])->name('alat.create');
    Route::post('/alat', [AdminController::class, 'storeAlat'])->name('alat.store');
    Route::get('/alat/{id}/edit', [AdminController::class, 'editAlat'])->name('alat.edit');
    Route::put('/alat/{id}', [AdminController::class, 'updateAlat'])->name('alat.update');
    Route::delete('/alat/{id}', [AdminController::class, 'destroyAlat'])->name('alat.destroy');
});

```

4. Tampilan akan seperti dibawah ini

PANEL ADMIN

[Dashboard](#)

[Kelola User](#)

[Kelola Kategori](#)

[Kelola Alat](#)

Manajemen Data Alat

Logout

Data alat berhasil diperbarui.

Daftar Alat Laboratorium

Cari

+ Tambah Alat

GAMBAR	NAMA ALAT	KATEGORI	STOK	KONDISI	AKSI
	infocus	Alat Elektronik	1	Baik	Edit Hapus

PANEL ADMIN

[Dashboard](#)

[Kelola User](#)

[Kelola Kategori](#)

[Kelola Alat](#)

Tambah Alat Baru

Logout

Nama Alat

Kategori

-- Pilih Kategori --

Stok

1

Status Kondisi

Baik

Deskripsi (Opsional)

Gambar Alat (Opsional)

Choose File

No file chosen

Batal

Simpan

PANEL ADMIN

[Dashboard](#)

[Kelola User](#)

[Kelola Kategori](#)

[Kelola Alat](#)

Edit Data Alat

Logout

Nama Alat

Kategori

Alat Elektronik

Stok

1

Status Kondisi

Baik

Deskripsi

Gambar Alat (Biarkan kosong jika tidak ingin mengubah gambar)



Choose File

No file chosen

Batal

Perbarui

MEMBUAT CRUD PEMINJAMAN

Untuk membuat fitur CRUD Peminjaman ini, kita harus memperhitungkan relasi Master-Detail (antara tabel peminjamans dan detail_pinjams). Artinya, dalam satu transaksi peminjaman, pengguna bisa meminjam satu atau lebih jenis alat dengan jumlah tertentu.

Berikut adalah langkah-langkah lengkap untuk mengimplementasikannya ke dalam Controller, Route, dan Views

Langkah 1: Tambahkan Method Peminjaman di AdminController.php

Tambahkan logika untuk menampilkan daftar peminjaman, form tambah, proses simpan (yang otomatis mengurangi stok alat), detail peminjaman, update status, dan penghapusan data:

```
// 1. Menampilkan daftar peminjaman
1 reference | 0 overrides
public function indexPeminjaman(Request $request)
{
    $search = $request->input('search');

    $peminjamans = Peminjaman::with(['user', 'detailPinjams.alat'])
        ->when($search, function ($query, $search) {
            return $query->where('status', 'like', "%{$search}%")
                ->orWhereHas('user', function ($q) use ($search) {
                    $q->where('name', 'like', "%{$search}%");
                });
        })
        ->latest()
        ->paginate(10)
        ->withQueryString();

    return view('admin.peminjaman.index', compact('peminjamans', 'search'));
}

// 2. Menampilkan form tambah peminjaman
1 reference | 0 overrides
public function createPeminjaman()
{
    $users = User::where('role', 'peminjam')->get(); // Atau ambil semua user jika bebas
    $alats = Alat::where('stok', '>', 0)->get();
    return view('admin.peminjaman.create', compact('users', 'alats'));
}
```

```
// 3. Menyimpan data peminjaman baru
```

```
1 reference | 0 overrides
```

```
public function storePeminjaman(Request $request)
```

```
{
```

```
    $request->validate([
```

```
        'user_id' => 'required|exists:users,id',
```

```
        'tgl_pinjam' => 'required|date',
```

```
        'tgl_kembali_plan' => 'required|date|after_or_equal:tgl_pinjam',
```

```
        'alat_id' => 'required|array',
```

```
        'alat_id.*' => 'exists:alats,id',
```

```
        'jumlah' => 'required|array',
```

```
        'jumlah.*' => 'integer|min:1',
```

```
    ]);
```

```
    DB::beginTransaction();
```

```
    try {
```

```
        // Buat transaksi utama peminjaman
```

```
        $peminjaman = Peminjaman::create([
```

```
            'user_id' => $request->user_id,
```

```
            'tgl_pinjam' => $request->tgl_pinjam,
```

```
            'tgl_kembali_plan' => $request->tgl_kembali_plan,
```

```
            'status' => 'diajukan', // Status awal
```

```
        ]);
```

```
        // Simpan detail alat yang dipinjam
```

```
        foreach ($request->alat_id as $index => $alatId) {
```

```
            $jumlahPinjam = $request->jumlah[$index];
```

```
            $alat = Alat::findOrFail($alatId);
```

```
            // Validasi stok
```

```
            if ($alat->stok < $jumlahPinjam) {
```

```
                throw new \Exception("Stok alat '{$alat->nama_alat}' tidak mencukupi.");
```

```
            }
```

```
        DetailPinjam::create([
```

```
            'peminjaman_id' => $peminjaman->id,
```

```
            'alat_id' => $alatId,
```

```
            'jumlah' => $jumlahPinjam,
```

```
        ]);
```

```
        // Kurangi stok alat jika status langsung disetujui/dipinjam (Opsional, atau dikurangi saat status berubah jadi 'dipinjam')
```

```
    }
```

```
    DB::commit();
```

```
    return redirect()->route('admin.peminjaman.index')->with('success', 'Data peminjaman berhasil diajukan.');
```

```
} catch (\Exception $e) {
```

```
    DB::rollBack();
```

```
    return back()->withInput()->with('error', $e->getMessage());
```

```
}
```

```
}
```

```
// 4. Memperbarui status peminjaman (Misal: dari diajukan -> dipinjam / selesai)
1 reference | 0 overrides
public function updateStatusPeminjaman(Request $request, $id)
{
    $peminjaman = Peminjaman::with('detailPinjams.alat')->findOrFail($id);

    $request->validate([
        'status' => 'required|in:diajukan,dipinjam,selesai,telat',
    ]);

    DB::beginTransaction();
    try {
        $statusLama = $peminjaman->status;
        $statusBaru = $request->status;

        // Logika pengelolaan stok otomatis
        if ($statusLama != 'dipinjam' && $statusBaru == 'dipinjam') {
            // Kurangi stok karena barang resmi dipinjam
            foreach ($peminjaman->detailPinjams as $detail) {
                $alat = $detail->alat;
                if ($alat->stok < $detail->jumlah) {
                    throw new \Exception("Stok alat {$alat->nama_alat} tidak mencukupi untuk dipinjam.");
                }
                $alat->decrement('stok', $detail->jumlah);
            }
        } elseif ($statusLama == 'dipinjam' && ($statusBaru == 'selesai')) {
            // Kembalikan stok karena barang sudah dikembalikan (selesai)
            foreach ($peminjaman->detailPinjams as $detail) {
                $detail->alat->increment('stok', $detail->jumlah);
            }
        }

        $peminjaman->update(['status' => $statusBaru]);
    }
}
```

```
        DB::commit();
        return redirect()->route('admin.peminjaman.index')->with('success', 'Status peminjaman berhasil diperbarui.');
```

```
} catch (\Exception $e) {
    DB::rollBack();
    return back()->with('error', $e->getMessage());
}
}
```

```
// 5. Menghapus data peminjaman
1 reference | 0 overrides
public function destroyPeminjaman($id)
{
    $peminjaman = Peminjaman::with('detailPinjams')->findOrFail($id);

    // Jika statusnya sedang dipinjam, kembalikan stok terlebih dahulu sebelum dihapus
    if ($peminjaman->status == 'dipinjam') {
        foreach ($peminjaman->detailPinjams as $detail) {
            $detail->alat->increment('stok', $detail->jumlah);
        }
    }

    $peminjaman->delete();

    return redirect()->route('admin.peminjaman.index')->with('success', 'Data peminjaman berhasil dihapus.');
```

```
}
```

Langkah 2: Daftarkan Route di routes/web.php

Tambahkan rute-rute berikut ke dalam kelompok middleware admin:

```
//CRUD Peminjaman
Route::get('/peminjaman', [AdminController::class, 'indexPeminjaman'])->name('peminjaman.index');
Route::get('/peminjaman/create', [AdminController::class, 'createPeminjaman'])->name('peminjaman.create');
Route::post('/peminjaman', [AdminController::class, 'storePeminjaman'])->name('peminjaman.store');
Route::put('/peminjaman/{id}/status', [AdminController::class, 'updateStatusPeminjaman'])->name('peminjaman.updateStatus');
Route::delete('/peminjaman/{id}', [AdminController::class, 'destroyPeminjaman'])->name('peminjaman.destroy');
```

Langkah 3: Buat Tampilan Views Peminjaman

Buat folder baru bernama peminjaman di dalam resources/views/admin/, lalu buat 2 file view berikut (index.blade.php dan create.blade.php).

A. resources/views/admin/peminjaman/index.blade.php

```
@extends('layouts.app')

@section('title', 'Kelola Peminjaman - Panel Admin')
@section('header-title', 'Manajemen Transaksi Peminjaman')

@section('content')
    @if(session('success'))
        <div class="mb-4 bg-emerald-50 border border-emerald-200 text-emerald-800 p-4 rounded-lg shadow-sm text-sm">
            {{ session('success') }}
        </div>
    @endif
    @if(session('error'))
        <div class="mb-4 bg-red-50 border border-red-200 text-red-800 p-4 rounded-lg shadow-sm text-sm">
            {{ session('error') }}
        </div>
    @endif

    <div class="bg-white rounded-lg shadow-sm overflow-hidden border border-gray-200">
        <div class="p-5 border-b border-gray-200 bg-gray-50 flex flex-col md:flex-row justify-between items-center gap-4">
            <h3 class="text-lg font-bold text-gray-800">Daftar Transaksi Peminjaman</h3>

            <div class="flex items-center gap-3 w-full md:w-auto">
                <!-- Form Search -->
                <form action="{{ route('admin.peminjaman.index') }}" method="GET" class="flex w-full md:w-80">
                    <input type="text" name="search" value="{{ request('search') }}" placeholder="Cari nama peminjam / status..."
                        class="w-full px-3 py-2 text-sm border border-gray-300 rounded-l-lg focus:outline-none focus:ring-2 focus:ring-blue-500">
                    <button type="submit" class="bg-gray-800 hover:bg-gray-900 text-white px-4 py-2 text-sm font-semibold rounded-r-lg transition">
                        Cari
                    </button>
                </form>
                @if(request('search'))
                    <a href="{{ route('admin.peminjaman.index') }}"
                        class="ml-2 bg-gray-300 hover:bg-gray-400 text-gray-700 px-3 py-2 text-sm rounded-lg flex items-center transition">
                        Reset
                    </a>
                @endif
            </div>
        </div>
    </div>
</form>
```

```
<!-- Tombol Tambah -->
<a href="{{ route('admin.peminjaman.create') }}"
    class="bg-blue-600 hover:bg-blue-700 text-white text-sm font-semibold px-4 py-2 rounded-lg transition whitespace-nowrap">
    + Tambah Peminjaman
</a>
</div>
</div>

<div class="overflow-x-auto">
    <table class="w-full text-left border-collapse">
        <thead>
            <tr class="bg-gray-100 text-gray-600 text-sm uppercase tracking-wider">
                <th class="py-3 px-4 border-b">Peminjam</th>
                <th class="py-3 px-4 border-b">Alat yang Dipinjam</th>
                <th class="py-3 px-4 border-b">Tgl Pinjam / Rencana Kembali</th>
                <th class="py-3 px-4 border-b">Status</th>
                <th class="py-3 px-4 border-b">Aksi</th>
            </tr>
        </thead>
    </table>
</div>
```

```

        </tr>
    </thead>
    <tbody class="text-gray-700 text-sm">
        @forelse($peminjamans as $peminjaman)
            <tr class="hover:bg-gray-50 transition align-top">
                <td class="py-3 px-4 border-b font-medium text-gray-900">
                    {{ $peminjaman->user->name ?? 'User Dihapus' }}
                </td>
                <td class="py-3 px-4 border-b">
                    <ul class="list-disc list-inside space-y-1">
                        @foreach($peminjaman->detailPinjams as $detail)
                            <li>
                                <span class="font-semibold">{{ $detail->alat->nama_alat ?? 'Alat Dihapus' }}</span>
                                <span class="text-xs bg-gray-200 px-1.5 py-0.5 rounded">{{ $detail->jumlah }} pcs</span>
                            </li>
                        @endforeach
                    </ul>
                </td>
            </tr>
        </tbody>
    </table>
    <div class="py-3 px-4 border-b text-xs text-gray-600">
        <span class="block">Pinjam: {{ $peminjaman->tgl_pinjam }}</span>
        <span class="block font-semibold">Rencana: {{ $peminjaman->tgl_kembali_plan }}</span>
    </div>
    <td class="py-3 px-4 border-b">
        <span class="px-2.5 py-1 text-xs font-semibold rounded-full">
            @if($peminjaman->status == 'diajukan') bg-yellow-100 text-yellow-800
            @elseif($peminjaman->status == 'dipinjam') bg-blue-100 text-blue-800
            @elseif($peminjaman->status == 'selesai') bg-emerald-100 text-emerald-800
            @else bg-red-100 text-red-800 @endif">
                {{ ucfirst($peminjaman->status) }}
            </span>
    </td>
    <td class="py-3 px-4 border-b">
        <div class="flex flex-col space-y-2">
            <!-- Form Ubah Status Cepat -->
            <form action="{{ route('admin.peminjaman.updateStatus', $peminjaman->id) }}" method="POST" class="flex items-center space-x-1">
                @csrf
                @method('PUT')
                <select name="status" onchange="this.form.submit()" class="text-xs border border-gray-300 rounded px-2 py-1 focus:outline-none">
                    <option value="diajukan" {{ $peminjaman->status == 'diajukan' ? 'selected' : '' }}>Diajukan</option>
                    <option value="dipinjam" {{ $peminjaman->status == 'dipinjam' ? 'selected' : '' }}>Dipinjam</option>
                    <option value="selesai" {{ $peminjaman->status == 'selesai' ? 'selected' : '' }}>Selesai</option>
                    <option value="telat" {{ $peminjaman->status == 'telat' ? 'selected' : '' }}>Telat</option>
                </select>
            </form>

            <!-- Tombol Hapus -->
            <form action="{{ route('admin.peminjaman.destroy', $peminjaman->id) }}" method="POST">
                @csrf
                @method('DELETE')
                <button type="submit" class="bg-red-500 hover:bg-red-600 text-white px-3 py-1 rounded text-xs font-semibold transition w-full">
                    Hapus
                </button>
            </form>
        </div>
    </td>
    </tr>
    @empty
        <tr>
            <td colspan="5" class="py-4 text-center text-gray-500">Belum ada data peminjaman.</td>
        </tr>
    @endforelse
    </tbody>
</table>
</div>

<div class="p-4 border-t border-gray-200 bg-gray-50">
    {{ $peminjamans->links() }}
</div>
</div>
</endsection>

```

B. resources/views/admin/peminjaman/create.blade.php

(Dilengkapi skrip sederhana agar admin bisa menambah baris alat dinamis jika ingin meminjam lebih dari 1 alat)

```
@extends('layouts.app')

@section('title', 'Tambah Peminjaman - Panel Admin')
@section('header-title', 'Form Tambah Transaksi Peminjaman')

@section('content')


@if(session('error'))
        <div class="mb-4 bg-red-50 border border-red-200 text-red-800 p-3 rounded-lg text-sm">
            {{ session('error') }}
        </div>
    @endif

    <form action="{{ route('admin.peminjaman.store') }}" method="POST">
        @csrf

        <div class="mb-4">
            <label class="block text-gray-700 text-sm font-semibold mb-2">Pilih Peminjam (User)</label>
            <select name="user_id" required class="w-full px-3 py-2 border border-gray-300 rounded-lg focus:outline-none focus:ring-2 focus:ring-blue-500">
                <option value=""--> Pilih User --</option>
                @foreach($users as $user)
                    <option value="{{ $user->id }}" {{ old('user_id') == $user->id ? 'selected' : '' }}>
                        {{ $user->name }} ({{ $user->email }})
                    </option>
                @endforeach
            </select>
        </div>

        <div class="grid grid-cols-1 md:grid-cols-2 gap-4 mb-4">
            <div>
                <label class="block text-gray-700 text-sm font-semibold mb-2">Tanggal Pinjam</label>
                <input type="date" name="tgl_pinjam" value="{{ old('tgl_pinjam', date('Y-m-d')) }}" required
                    class="w-full px-3 py-2 border border-gray-300 rounded-lg focus:outline-none focus:ring-2 focus:ring-blue-500">
            </div>
            <div>
                <label class="block text-gray-700 text-sm font-semibold mb-2">Rencana Tanggal Kembali</label>
                <input type="date" name="tgl_kembali_plan" value="{{ old('tgl_kembali_plan', date('Y-m-d', strtotime('+3 days')) }}" required
                    class="w-full px-3 py-2 border border-gray-300 rounded-lg focus:outline-none focus:ring-2 focus:ring-blue-500">
            </div>
        </div>

        <!-- Bagian Daftar Alat yang Dipinjam (Dinamis) -->
        <div class="mb-6">
            <label class="block text-gray-700 text-sm font-semibold mb-2">Daftar Alat yang Dipinjam</label>
            <div id="alat-container" class="space-y-3">
                <div class="flex items-center gap-2 alat-row">
                    <select name="alat_id[]" required class="flex-1 px-3 py-2 border border-gray-300 rounded-lg text-sm focus:outline-none">
                        <option value=""--> Pilih Alat --</option>
                        @foreach($alats as $alat)
                            <option value="{{ $alat->id }}">{{ $alat->nama_alat }} (Stok: {{ $alat->stok }})</option>
                        @endforeach
                    </select>
                    <input type="number" name="jumlah[]" value="1" min="1" placeholder="Jumlah" required
                        class="w-24 px-3 py-2 border border-gray-300 rounded-lg text-sm focus:outline-none">
                    <button type="button" onclick="removeRow(this)" class="bg-red-500 text-white px-3 py-2 rounded-lg text-sm hover:bg-red-600 transition">X</button>
                </div>
                <div>
                    <button type="button" onclick="addRow()" class="mt-3 bg-gray-800 hover:bg-gray-900 text-white text-xs font-semibold px-3 py-2 rounded-lg transition">
                        + Tambah Alat Lain
                    </button>
                </div>
            </div>

            <div class="flex justify-end space-x-2">
                <a href="{{ route('admin.peminjaman.index') }}"
                    class="bg-gray-300 hover:bg-gray-400 text-gray-800 px-4 py-2 rounded-lg text-sm font-semibold transition">Batal</a>
                <button type="submit" class="bg-blue-600 hover:bg-blue-700 text-white px-4 py-2 rounded-lg text-sm font-semibold transition">Simpan Peminjaman</button>
            </div>
        </form>
    </div>

    <!-- Script Sederhana untuk Tambah/Hapus Baris Alat -->
    <script>
        function addRow() {
            const container = document.getElementById('alat-container');


```

```

const firstRow = container.querySelector('.alat-row');
const newRow = firstRow.cloneNode(true);
newRow.querySelector('select').value = '';
newRow.querySelector('input').value = '1';
container.appendChild(newRow);
}

function removeRow(button) {
    const rows = document.querySelectorAll('.alat-row');
    if (rows.length > 1) {
        button.closest('.alat-row').remove();
    } else {
        alert('Minimal harus ada 1 alat yang dipilih.');
```

Jangan lupa juga untuk menambahkan menu Kelola Peminjaman pada Sidebar di `resources/views/layouts/app.blade.php`:

```

<!-- Menu Kelola Peminjam -->
<a href="{{ route('admin.peminjaman.index') }}" class="block px-4 py-2 rounded-lg transition
{{ request()->routeIs('admin.peminjaman*') ? 'bg-gray-800 text-white font-medium shadow' : 'text-gray-400 hover:bg-gray-800 hover:text-white' }}">
Kelola Peminjaman</a>
```

Tampilannya akan seperti ini :

Halaman Utama Peminjaman (admin)

PANEL ADMIN

Dashboard

Kelola User

Kelola Kategori

Kelola Alat

Kelola Peminjaman

Manajemen Transaksi Peminjaman

Logout

Daftar Transaksi Peminjaman

Cari nama peminjam / status...

Cari

+ Tambah Peminjaman

PEMINJAM	ALAT YANG DIPINJAM	TGL PINJAM / RENCANA KEMBALI	STATUS	AKSI
Fajar Oktavian	• infocus (1 pcs)	Pinjam: 2026-08-09 Rencana: 2026-08-12	Diajukan	Diajukan Hapus

Halaman Tambah Peminjaman (admin)

Form Tambah Transaksi Peminjaman

Pilih Peminjam (User)

Fajar Oktavian (peminjam@app.com)

Tanggal Pinjam

09/08/2026

Rencana Tanggal Kembali

12/08/2026

Daftar Alat yang Dipinjam

Printer (Stok: 2)

1

X

+ Tambah Alat Lain

Batal

Simpan Peminjaman

Mengubah Status Pemijaman (admin)

Manajemen Transaksi Peminjaman

Logout

Data peminjaman berhasil diajukan.

Daftar Transaksi Peminjaman

Cari nama peminjam / status...

Cari

+ Tambah Peminjaman

PEMINJAM	ALAT YANG DIPINJAM	TGL PINJAM / RENCANA KEMBALI	STATUS	AKSI
Fajar Oktavian	• Printer (1 pcs)	Pinjam: 2026-08-09 Rencana: 2026-08-12	Diajukan	✓ Diajukan Dipinjam Selesai Telat Hapus
Fajar Oktavian	• infocus (1 pcs)	Pinjam: 2026-08-09 Rencana: 2026-08-12	Diajukan	